ECCC Movebank Manual

For GPS archival & base station, GSM, and satellite trackers

Updated: 12 June 2026

This document outlines a standardized approach to archiving seabird tracking data collected by Environment and Climate Change Canada (ECCC). Applying this approach will improve data discoverability, useability, and security. It was developed by and for the Biodiversity Research Division (formerly Wildlife Research Division) under the Integrated Marine Response Planning initiative of the Oceans Protection Plan. We encourage Canadian Wildlife Service researchers to consider adopting this approach as well.

In addition to supporting ECCC Open Data initiatives and serving as a secure back-up of data collected by ECCC, Movebank allows different levels of data access for the public, collaborators, and data managers – a feature that can be used to grant other ECCC researchers download access to the most current version of your dataset through the 'move' R package. Multi-colony and multi-species analyses are becoming more common and are important for understanding seabird habitat use and vulnerability to threats at regional and national scales, as well as for identifying knowledge gaps. Standardization of data fields and units significantly improves dataset usability and efficiency, enabling such multi-study, large-scale analyses.

Finally, Movebank includes a free, formal archiving service. They will work with you to publish only the data that you used in your analysis, carefully curating it and providing you with a doi to cite in your peer-reviewed publication. This ensures reproducible research and allows your original Movebank study to continue to evolve over time.

There are two companion documents to this manual: 1) an **ECCC Movebank Attribute Dictionary**, and 2) a **Reference Data Template** that can be used if prefer not to build your own reference data file in R based on the Attribute Dictionary fields.

We welcome feedback to improve these standards and tools! Please email your suggestions to WRD Tracking East and West to ensure a response.

Movebank study overview

When possible, there should be a separate Movebank study for each unique combination of species, basic tag type (GPS, GLS), and deployment location, but multiple years of data should be maintained within the same study. Given that we often work with different collaborators at different deployment locations, splitting studies based on this factor will allow us greater control over data sharing permissions. Ideally, Movebank studies should be created prior to device deployment (required for GSM or satellite feeds), and should be updated once a year upon completion of the field season. If you are using archival or base station tags, you can create your study following the field season, but please review the required data fields in the ECCC Movebank Attribute Dictionary prior to field work!

All Movebank studies have four main components:

1) study details,

2) reference data (Movebank's term for metadata, includes animal, tag, and deployment attributes),

3) event-level data (Movebank's term for tag data, includes positional data and other accessory data), and

4) attachments (copies of original reference and event-level data, plus optional supporting files).

Study details are shown on the landing page of the study, reference data and event-level data are imported into Movebank's database and can be accessed programmatically, and attachments can only be accessed by manual download from the study page.

Movebank only accepts specific attributes, or 'fields', for import into their database from your reference and event-level datasets (more on this in section 'Prepare your data for Movebank'). However, the files you upload can include fields that Movebank's database does not accept. These files will automatically be saved, as-is, to the attachments section of the study. Because of this default file-archiving behaviour, we suggest you name each data file something meaningful before you upload it. Here are some examples:

- Reference data file
  'referenceData'_spcd_tagType_site.csv
- GPS positional data file
  'posData'_spcd_tagType_site.csv
- TDR depth/pressure data file

'tdrData'_spcd_tagType_site.csv

Importing data to Movebank's database is easier if you first reformat the original versions of your reference data and the raw data you get from your tags so that the field names and values (units) match those accepted by the system. You may also wish to identify outliers in your tracking data. For Movebank to be effective as a data backup system and to make your data processing reproducible and revisable, all files required to complete these processes should be included in the attachments section of the study as a zip file. We provide a suggested approach using a minimal R project where you can generate a zip file of just your script(s) and the raw input data in section 'Prepare your data for Movebank'. The contents of this folder can be organized however you like, as long as it works and is easily interpreted by a third party. Please include the date in this filename so it is clear when the data were last updated.

- Zipped folder containing original data files and file processing scripts
  'MovebankDataPrep'_spcd_tagType_site_yyyymmdd.zip

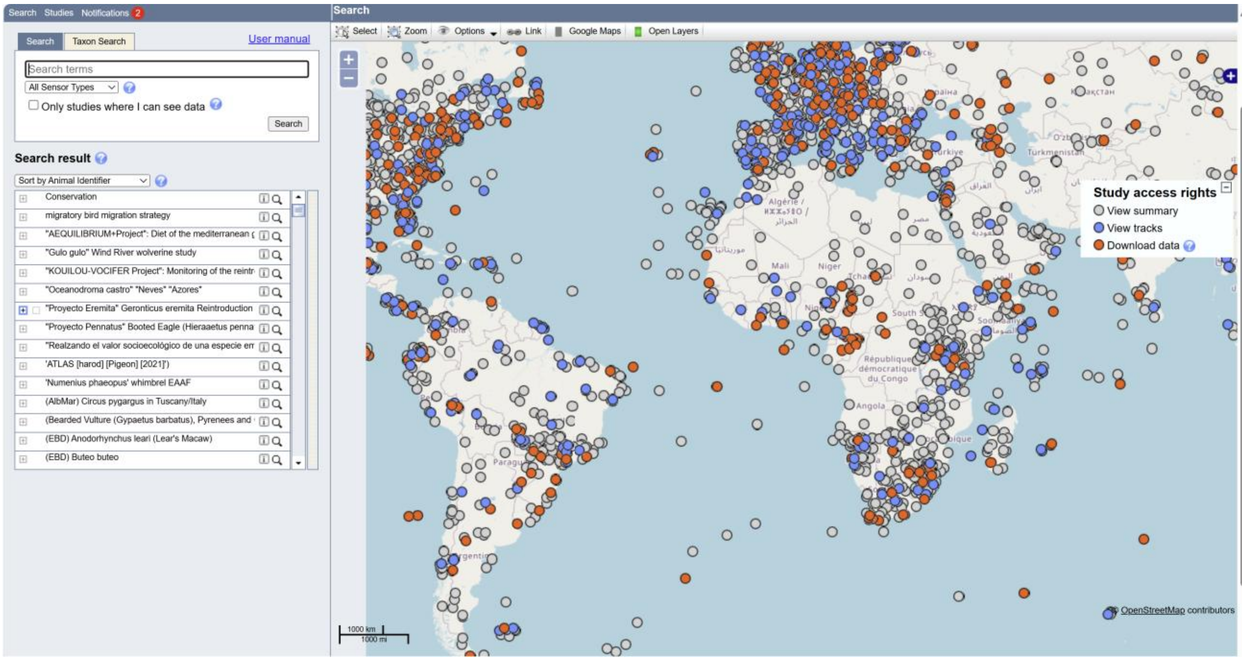
Finally, we request that you include a document that briefly outlines any changes that still need to be made to the study (e.g. missing data), any important processing steps (e.g. how outlier positions were identified), or any issues that someone using the data would want to know about (e.g. gappy tracks in a particular year). We intend to provide an R Markdown template to help generate this document in the 'seamoveR' R package, but for now, a simple text file can be used if needed.

- User guidance file
  README.txt

Initiate your Movebank study

Create a study

When you log into Movebank, you should see a landing page like the following:



Select the 'Studies' tab, then click the 'New Study' button at the top of the lefthand pane.

In the window that appears, you will start filling out the Study Attributes (see the ECCC Movebank Attribute Dictionary for details on all required fields and values).

Enter a standardized 'Study Name'. Please use the format:

> SpeciesName BasicTagType – LeadOrganization – StudySite, Region
>
> (e.g. 'Leach's storm-petrel GPS - ECCC - Baccalieu Is., Atlantic Canada')

Enter a Contact Person. This person must have a Movebank account. You populate this field by clicking 'Select', entering a Movebank username in the 'Filter' field, and then clicking it. This should be either your account (the uploader) or that of the project PI.

Enter a PI. Use the same 'Select' and search approach as above. This should be the current, or most recent, PI for the study. If the PI does not have a Movebank account, there is a box to click below the field that will allow you to enter PI contact information manually.

Enter Citation, Acknowledgements, and Grants Used to the best of your ability. You can update these as your study progresses.

Enter a Study Summary briefly summarizing the study objectives, methods, and results (if applicable). Permits and ethical approvals used during the completion of this work should also be identified. You can update this as your study progresses.

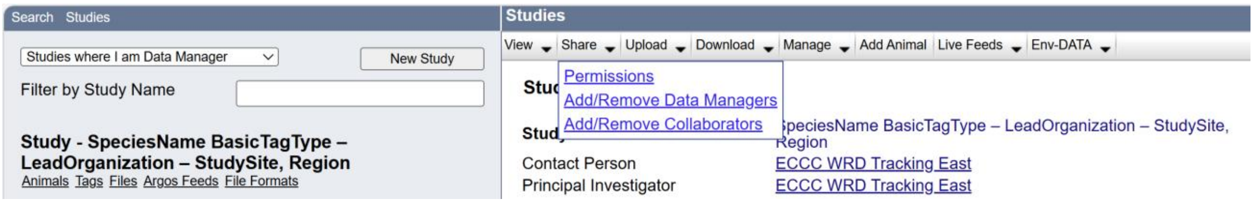
Enter Longitude and Latitude of the deployment site.

Set study type to 'Research'.

Click 'Save'.

Set study permissions

Once you have clicked save, you will be brought to a Permissions page. This page can also be accessed and revised later via the 'Share' dropdown menu.

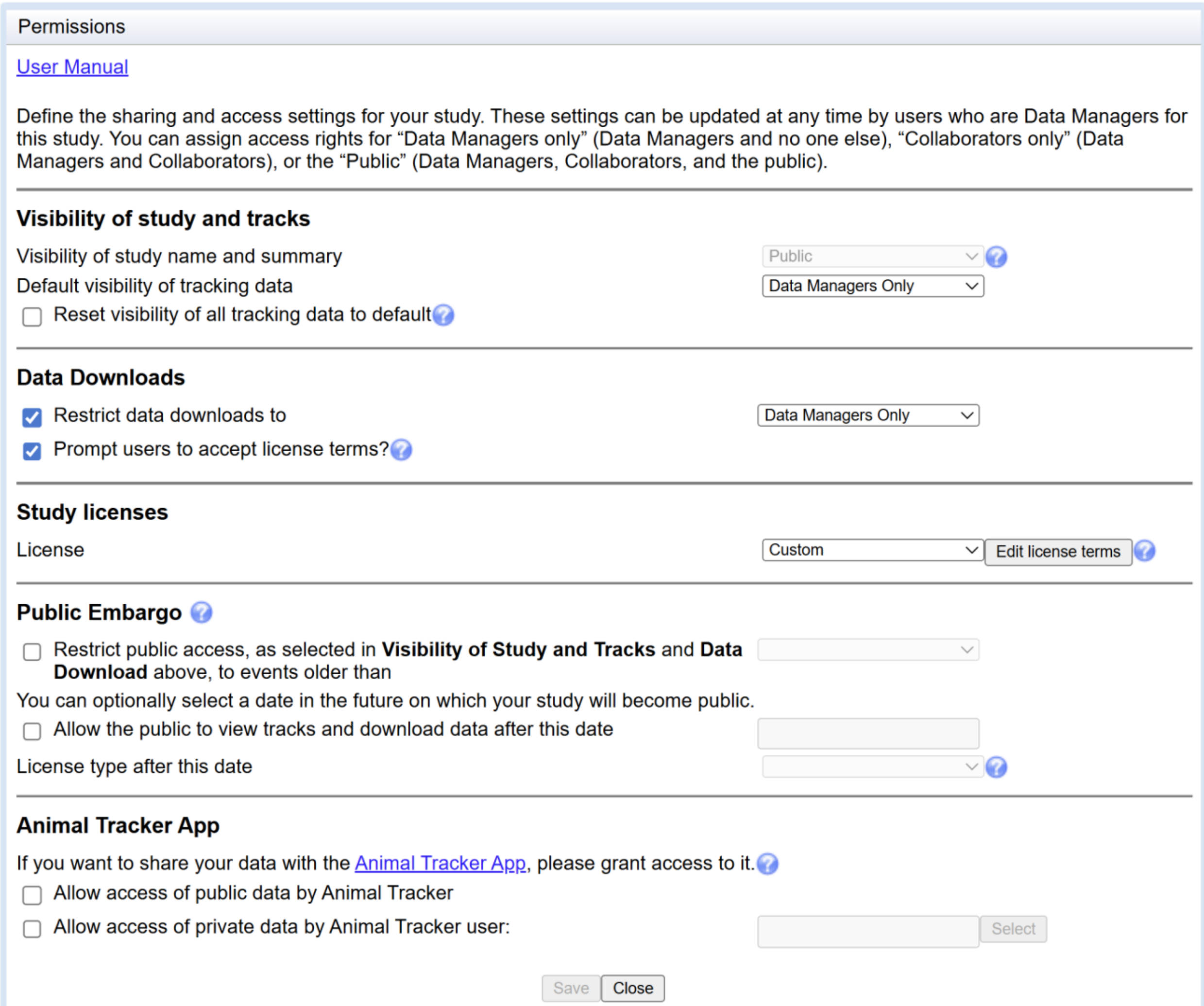


By creating the study, you are automatically designated as a 'Data Manager' with full access to data and administrative controls. You can designate additional people with Movebank accounts as 'Data Managers' or as 'Collaborators' (without administrative control over the study, but with customizable levels of access), through the 'Share' dropdown menu. All other users are considered 'Public'.

Until a study is truly finalized, we recommend that you set the 'Visibility of study name and summary' to 'Public', but the 'Default visibility of tracking data' to 'Data Managers only' or 'Collaborators only' (both collaborators and data managers).

Click the box for 'Restrict data downloads to' and set that to 'Data Managers only' or 'Collaborators only' as well. The other fields can be left as-is.

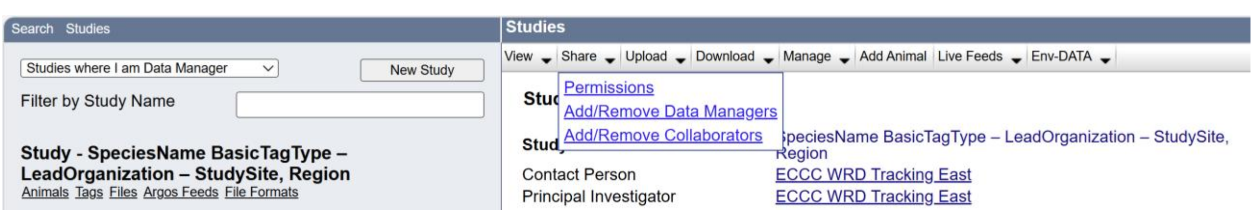
Click 'Save' and 'Close'.



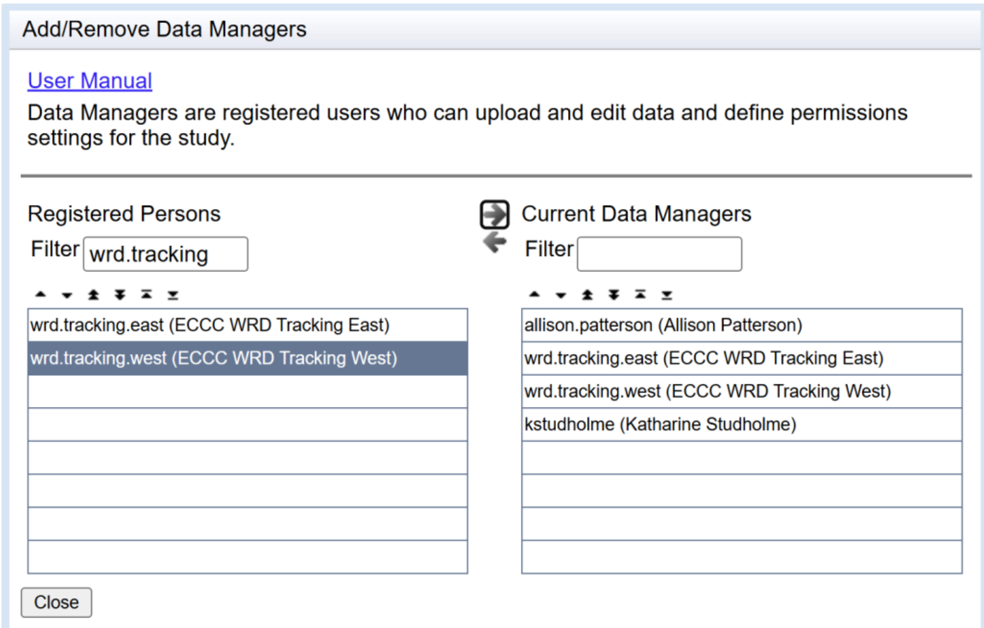
Add ECCC tracking accounts as Data Managers

If your study is led by ECCC, please add the ECCC Biodiversity Research Division (formerly WRD) Movebank accounts as Data Managers to enable study inventory, support, and data access. Projects led outside of ECCC should contact both their ECCC collaborators and the generic email for either WRD Tracking East or West to discuss what level of ECCC access would best support their goals.

From the study navigation bar in the right pane, select ‘Share’ -> ‘Add/Remove Data Managers’.



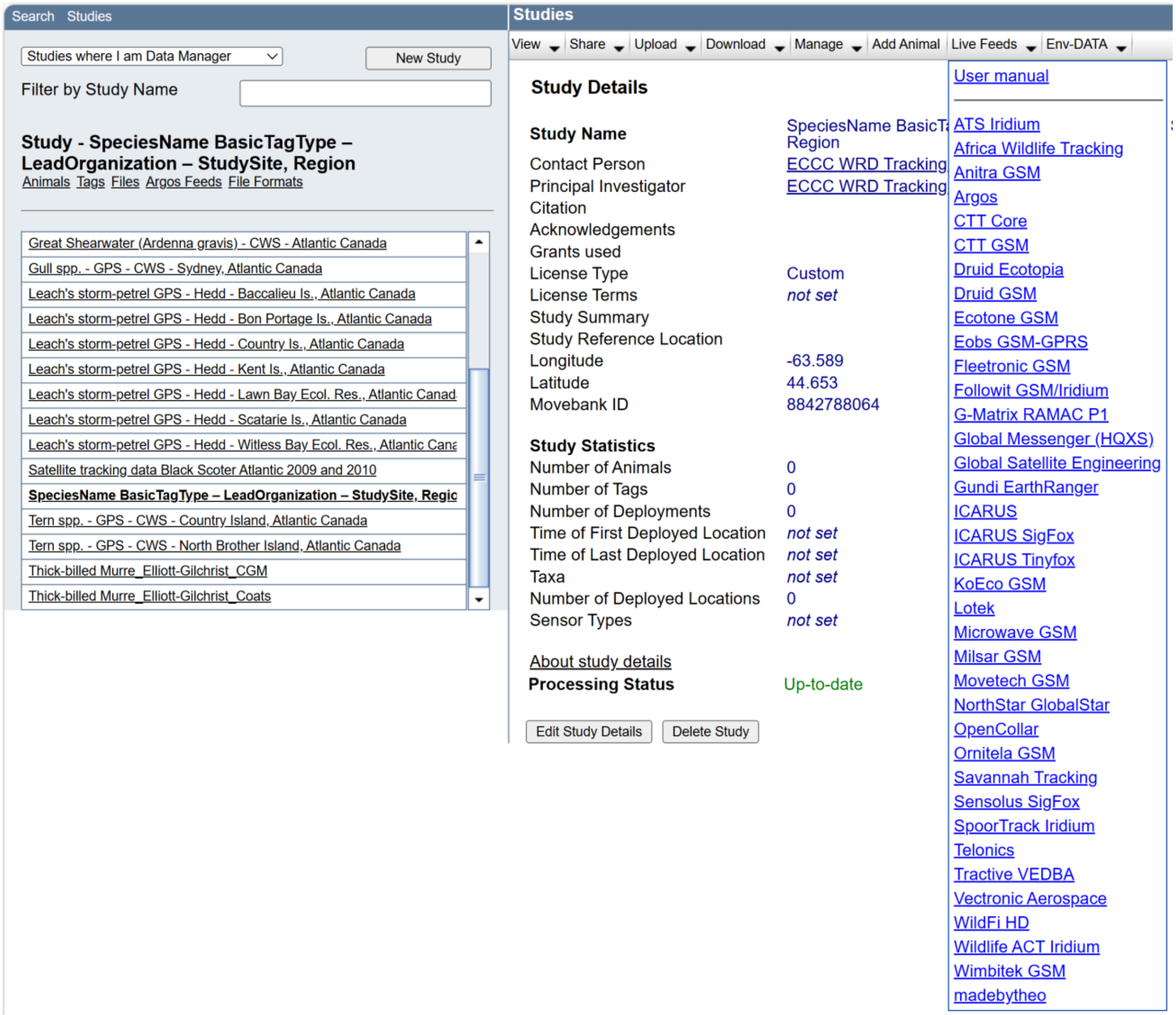
In the pop-up window, type ‘wrd.tracking’ into the ‘Filter’ field on the left-hand side under ‘Registered persons’. Accounts matching your search will appear below. Click on the accounts you wish to add (‘ECCC WRD Tracking East’ and ‘ECCC WRD Tracking West’), then click the arrow pointing to the right. The accounts should now appear under the ‘Current Data Managers’ section on the right-hand side. Click ‘close’.



Link tags with live feeds (e.g. GSM, PTT)

If the tags you are deploying transmit data over GSM or satellite connection, you must create a Movebank study to receive the data and then work with your tag manufacturer to have the tag information sent to your Movebank account. This involves providing the manufacturer with your Movebank username and a list of the tags you want access to, typically via an online portal hosted by the manufacturer or simply via email. Please see Movebank’s ‘Create and manage live data feeds’ page to identify the instructions relevant to your tag manufacturer.

Once the manufacturer has confirmed they’ve sent your tag information to Movebank, return to your study page, select the ‘Live Feeds’ dropdown menu, and then select your tag manufacturer. Select ‘New’, choose ‘Add all tags’ or choose individual tag IDs, then click the right arrow so the chosen tag IDs appear under ‘Selected tags’. Click ‘Save’.



The important next step is to **upload your reference data** according to the instructions in the following sections ‘Prepare your data for Movebank’ and ‘Upload and import data to Movebank’. Until you import reference data that links tag IDs to animal IDs and deployment periods, they will not be included in Movebank’s tracking data map or data downloads.

If you are using a tag type that transmits event-level data to base stations as well as over GSM connection, the following sections will also be relevant to you for handling those data. While in theory, event-level data sent to a base station should not also be reported over the data feed, this has been known to occur, resulting in duplicate records. We are currently working to develop guidance for GSM - base station data management based on user experiences this year. Please reach out if you would like to contribute your experiences to this effort!

IMPORTANT: Not all data associated with your tags will feed into your Movebank account! This service is for location data only. In addition to setting up your Movebank feed, you must also **download your data regularly from your tag manufacturer's data delivery platform**. Data hosted on these platforms typically expires after a set period of time – please clarify this with your manufacturer when planning your download schedule. To take full advantage of Movebank as a data archiving option, we recommend that you include these additional data in the Attachments section of your study (e.g. within a zipped folder containing original data files and file processing scripts – see following sections 'Prepare your data for Movebank' and 'Upload and import data to Movebank').

Please note: In 2025, Argos launched a new satellite system called Kineis. Data flowing through these satellites is accessible through a new data portal CLS - Land. Movebank does not yet have the ability to set up live feeds from this portal. If you are using Argos tags (e.g. Lotek Sunbird and PinPoint Argos), please contact your tag manufacturer directly to determine how to obtain and decompress/convert your data!

Prepare your data for Movebank

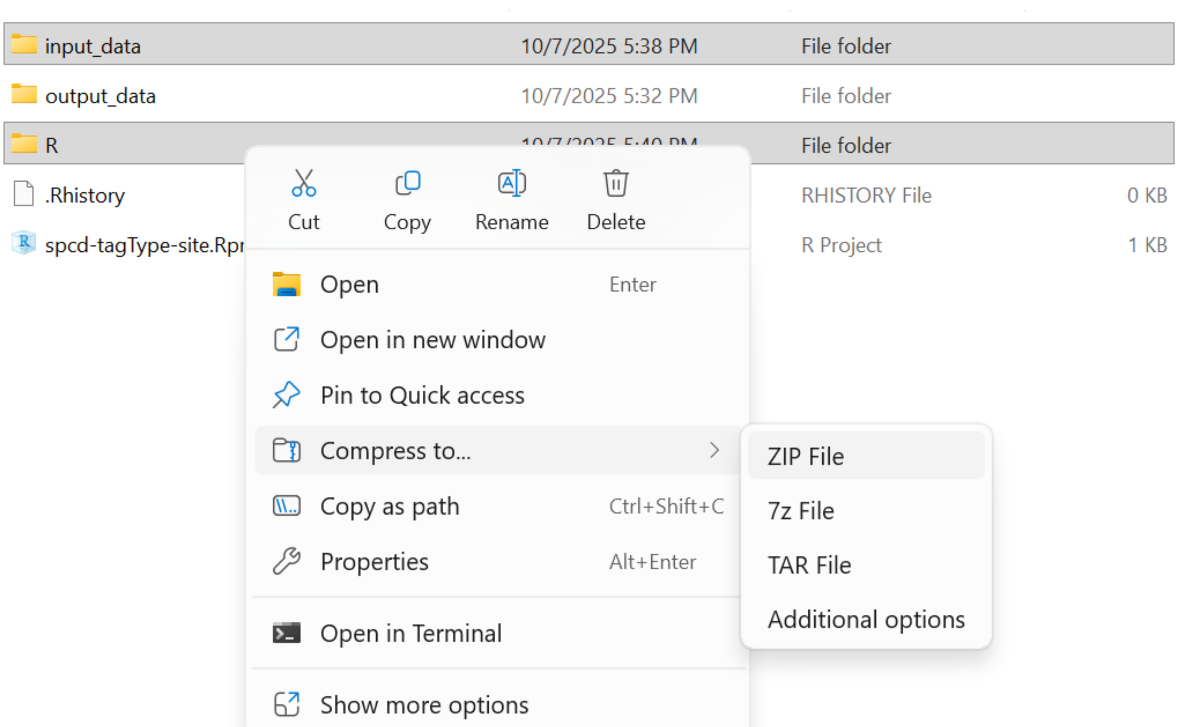
There are three main steps to preparing your data for Movebank upload and import to the database: 1) compile data from multiple tags into a single file for each data type (e.g. reference data, GPS positional data, TDR data), 2) reformat your data to have the appropriate fields, field names, and units, and 3) identify and flag (but don't remove!) erroneous positions in your tag data.

[If your tags are set up with a live feed to Movebank, you will not need to worry about these steps for your location data, but content related to reference data formatting and project organization are still relevant.]

We suggest that this process is best completed using an R project for each study and provide some helper functions and templates in R package 'seamoveR' (more on this below). Your R project can be organized however you like, as long as it contains everything needed to reproduce your work. Here is a suggested file structure to get you started:

spcd-tagType-site	Root folder
input_data	Folder for ‘raw’ input data, in subfolders
ref_data	Reference/metadata
event_data	Event data, in subfolders by manufacturer
basestation_data	Raw base station readouts subfolder
output_data	Folder for the reformatted files and README
R	Folder for R script(s)
spcd-tagType-site.Rproj	R project file

With your R project formatted in this way, you can easily select just the R scripts folder and the input_data folder, compress them to a zip file, give the zip file a meaningful name, and then upload it to the Movebank study as an Attachment to make your work fully reproducible. There is no need to include output_data or any other derived files in the zip file since all the material is there to re-create them. If you like, you can keep these zip files on your computer for version control or use a version control program like GitHub.



input_data	10/7/2025 5:38 PM	File folder	
output_data	10/7/2025 5:32 PM	File folder	
R	10/7/2025 5:40 PM	File folder	
.Rhistory	10/7/2025 5:40 PM	RHISTORY File	0 KB
spcd-tagType-site.Rproj	10/7/2025 5:38 PM	R Project	1 KB
MovebankDataPrep-spcd-tagType-site-20251007.zip	10/7/2025 9:55 PM	Compressed (zipped)...	2 KB

Reformat your data

Prior to upload, your data need to be consolidated into one file each for reference data, positional data, and any accessory data, using fields 'tag-ID', 'deploy-on-timestamp', and 'deploy-off-timestamp' in the reference data and fields for 'tag-ID' and 'timestamp' in the event-level data to differentiate each deployment (see the ECCC Movebank Attribute Dictionary). The consolidation of positional and accessory data should be done programmatically in R.

Once each data type is in one file (or R object), you will need to reformat your data to match Movebank's requirements and save the final versions as .csv files within the 'output_data' folder of your R project. Movebank has an extensive Attribute Dictionary that defines all the fields supported by their database (field names, standard units and categories). After a user uploads a file, they must assign, or 'map', their own data fields to the Movebank supported fields in order to import them into the database. While Movebank does provide tools to map fields with unsupported names and convert units and category definitions on the fly, we have found this to be a frustrating and error-prone process. **We recommend that you reformat your data prior to upload**.

When reformatting is done in advance, Movebank will recognize (most) field names in your file and suggest an appropriate match from its Attribute Dictionary, and you won't need to do any unit or category conversions on the fly, making the process much smoother.

Movebank's Attribute Dictionary contains far more fields than we will regularly need for ECCC studies. To simplify the reformatting process and standardize which fields are used, we have created a reduced '**ECCC Movebank Attribute Dictionary**' that defines required and optional fields to be used in ECCC studies. **Please use this document as your primary guide when reformatting your data**. Also, remember that you can upload a file containing additional fields that are not supported by Movebank if you wish – these fields will not be stored in the database, but will exist in the copy of your original file that is automatically saved in the Attachments section of the study. We suggest some optional 'custom fields' in the Deployment Attributes section of the ECCC Movebank Attribute Dictionary.

Reference data

Reproducibly reformatting your reference data in R, according to the ECCC Movebank Attribute Dictionary fields, is encouraged! Alternatively, reference data can be reformatted completing the **'Reference Data Template' excel file**, again referencing the Attribute Dictionary to understand Movebank field names and required units. Two fields are provided for datetime fields such that both local time (recorded in the field) and UTC (required by Movebank) are archived. To ensure a compatible format for Movebank in the new UTC fields, enter the datetime **as characters** by starting each cell with a single quotation mark ('), followed by the datetime in format yyyy-MM-dd HH:mm:ss.SSS. Further instructions on how to use the Reference Data Template are included in a README tab.

Fields 'tag ID' and 'animal ID' (typically the band number) should be used to generate a unique 'deployment ID'. Please be aware that if your study involves the deployment of the same tag on the same animal on more than one occasion this system will not result in a unique ID! In this case, you will need to add the deployment date, or some other differentiating value, to the deployment id.

Please retain all deployments in your reference data, even if the tag yielded no data.

Event-level data

The 'seamoveR' R package [devtools::install_github("krstudholme/seamoveR")] includes functions to compile and reformat event-level data from several tag types so they meet the requirements of Movebank. Functions are currently available for processing positional and TDR data from Ecotone, Pathtrack, and TechnoSmart GPS archival and base station tags. If there is no function available for the tag type you are using and you would like one added, or you are experiencing errors with existing functions, please email WRD Tracking East and West to ensure a response.

Regardless of whether you are using the seamoveR functions, event-level data should be reformatted programmatically in R. This will make it easier to consolidate and reformat multiple event-level files, filter duplicate records, update any errors (e.g. unit conversions!) quickly and easily, and seamlessly re-run code to append new years of data.

The functions in seamoveR do not clip your event-level data to the deployment period and we recommend that you also refrain from doing this. Movebank will automatically identify the deployment period based on the deploy-on and deploy-off UTC timestamps in your reference data and all original tag data will be retained in case of errors.

seamoveR functions also do not remove any event-level data other than duplicate records. Event-level data records where tags recorded NA,NA or 0,0 coordinate are instead marked as 'outliers' which hide them from the Movebank map and flag potential issues to future data users while retaining all original data in the database. In the case of Ecotone function, NA,NA records that occur within range of the base station can optionally be re-assigned to the base station coordinates instead of being marked as outliers.

If you choose to identify additional outlier positions (e.g. based on movement rate) after running the seamoveR formatting function, please make sure you change the 'import-marked-outlier' field to 'TRUE' for each new outlier record and add an explanation of your change and why you made it in the 'comments' field.

For burrow nesting species, you may also wish to identify cases of burrow use based on variables such as the number of satellites detected and distance of the previous or subsequent position from the colony. These records will have been marked as outliers by the seamoveR functions so, if you do this, you will need to change the 'import-marked-outlier' field to 'FALSE', replace the 0,0 coordinate with the deployment coordinate, and add an explanation of your change and why you made it in the 'comments' field.

All custom changes to event-level data should be implemented using an R script so they are reproducible. Such changes should also be outlined in the study's user guidance file (README file) to inform future data managers and analysts.

User guidance file

Please include a user guidance, or 'README' file, with your project. This document should give a brief summary of your study and outline any changes that still need to be made (e.g. missing data), any important pre-processing decisions you made (e.g. opting to identify cases of burrow use), and any issues that someone using the data would want to know about (e.g. gappy tracks in a particular year).

We plan to provide an R Markdown template within the seamoveR package to help generate a standardized .pdf version of this document, including basic study statistics, plots and maps. This tool is still in development, so for now please use a simple README.txt file.

Are you ready?

Before you import your data to Movebank, you should have:

- Reference data file
  'referenceData'_spcd_site_tagType.csv

- GPS positional data file [unless your positions are live-feed only]
  'posData'_spcd_tagType_site.csv

- Any other event-level data file(s)
  e.g. 'tdrData'_spcd_tagType_site.csv

- Zipped folder containing original data files and file processing scripts
  'MovebankDataPrep'_spcd_tagType_site_yyyymmdd.zip

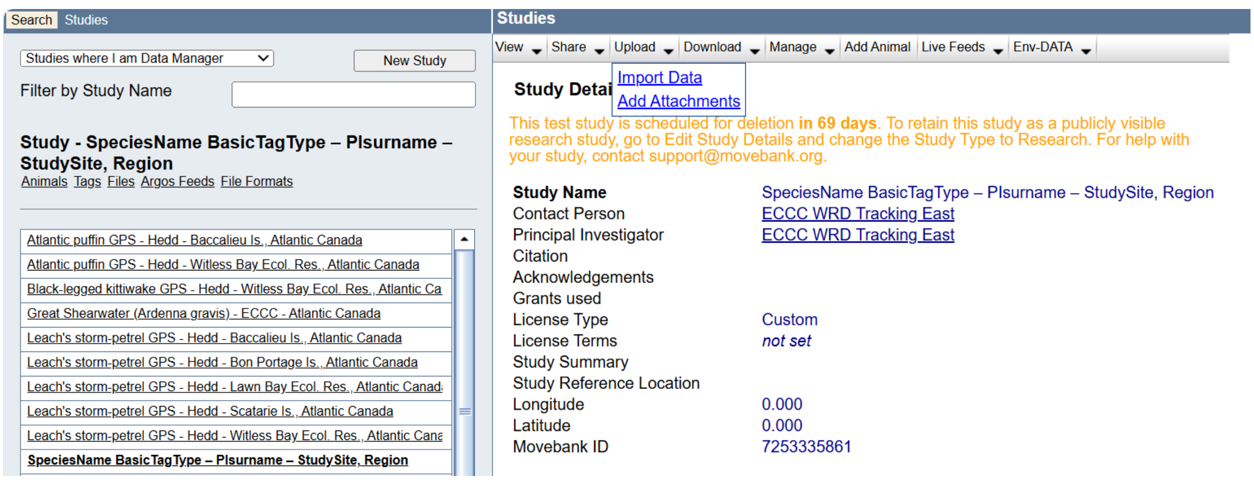
- User guidance file
  README.txt

Upload and import data to Movebank

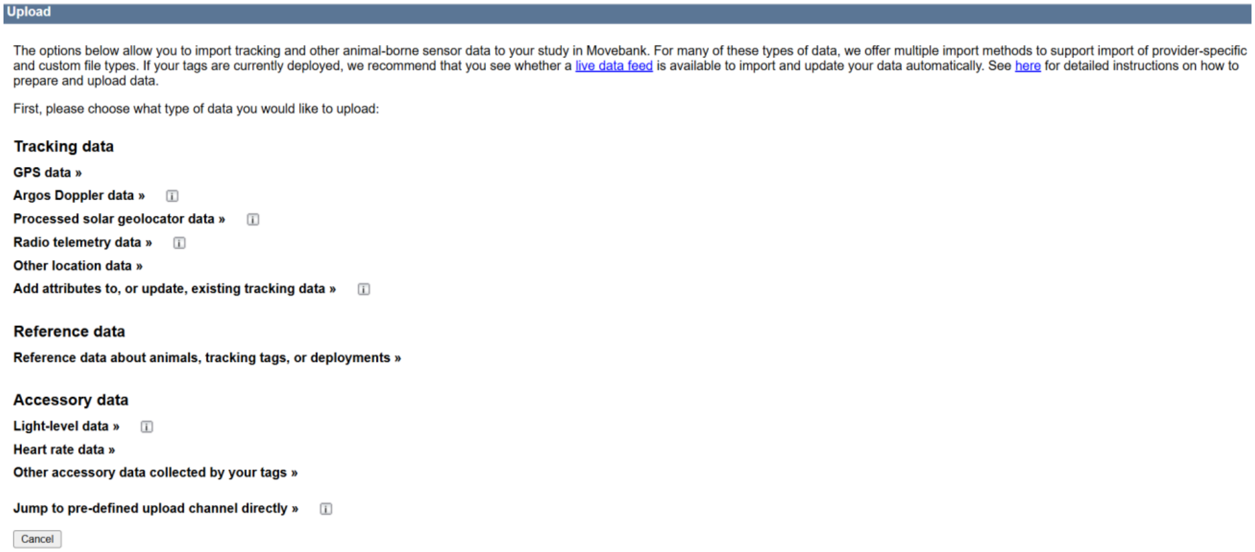
In the left-hand panel, make sure you're in the 'Studies' tab. (This resets to the 'Search' tab on login.)

Select the study you want to be working on in your Studies panel.

Then select 'Upload' -> 'Import Data'.



The following options will appear:



Upload

The options below allow you to import tracking and other animal-borne sensor data to your study in Movebank. For many of these types of data, we offer multiple import methods to support import of provider-specific and custom file types. If your tags are currently deployed, we recommend that you see whether a live data feed is available to import and update your data automatically. See here for detailed instructions on how to prepare and upload data.

First, please choose what type of data you would like to upload:

Tracking data

GPS data »

Argos Doppler data »

Processed solar geolocator data »

Radio telemetry data »

Other location data »

Add attributes to, or update, existing tracking data »

Reference data

Reference data about animals, tracking tags, or deployments »

Accessory data

Light-level data »

Heart rate data »

Other accessory data collected by your tags »

Jump to pre-defined upload channel directly »

Cancel

Reference data

In the 'Reference Data' section, select 'Reference data about animals, tracking tags, or deployments'

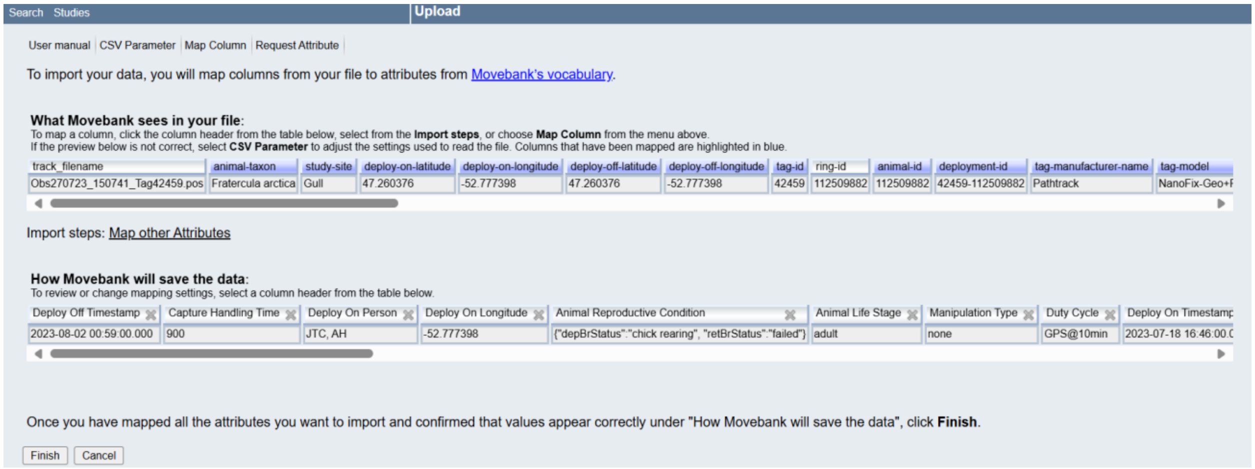
Select 'Use Movebank standard reference data format'

[If this is not your first time uploading reference data to this study and you saved a custom file format in the past (translation file to convert your data to Movebank's format), a popup window will offer you the chance to select it. You can do this or click 'skip'.]

Navigate to your file using the 'Choose file' button, select it, then hit the upload button

Click Ok to acknowledge the permissions popup window

The next page will show you two tables – the top table shows your dataset as uploaded and the bottom table shows your dataset as it will be saved in Movebank's database.



Columns that Movebank thinks it has successfully 'mapped' are highlighted in blue in the upper table and also appear in the lower table. Make sure the entries in the lower table have values, if you expect them to, and look for errors appearing in the 'Messages' Tab on the righthand side section of the screen. Check that the datetime columns are properly formatted (times are in UTC and match those in the upper table), that any categorical variables have all the categories you expect, and that latitude and longitude columns are not reversed. If you hover over a column in the upper table, it will highlight the corresponding column in the lower table to help you locate it.

If a field you expected to get mapped is grey, not blue, in the upper table, click on the field name and use the interactive window that appears to select the appropriate field name. Check that the units are correct, then hit 'Validate' and 'Save'. This should not happen often, but in the example above you can see that ring-id was not mapped properly. If you are successful in mapping the attribute, the column header will turn blue and the field will appear in the lower table.

Another possible issue is that a field can get mapped (appears blue in the upper table) but in the lower table the contents of the field are blank. This means Movebank recognized the field name but the format of the values did not match what Movebank expected. If this happens, you have two options: 1) Cancel the upload (hit the 'cancel' button, then in the popup 'Quit without save'), fix the formatting issue in your reference data file in R/by hand, and reupload the file, or 2) reformat the field within Movebank by clicking on the field name/heading in the lower table and using the popup window to tell Movebank how to interpret your values.

Unrecognized or incorrectly interpreted formats most often occur with datetime columns. Movebank expects datetimes to be formatted as 'yyyy-MM-dd HH:mm:ss.SSS'. Excel has a mind of its own and will often mess up this formatting, even if you ensured the data were entered as characters (use a single quotation mark before entering the cell contents). Just opening your reference data .csv before uploading can trigger reformatting! If you datetime is unrecognized, this is best resolved using option 1, above. But, if you're confident your datetimes are formatted correctly and you get an 'invalid dates' error when you click 'Finish' on your data upload, it's likely you didn't notice that Movebank interpreted your datetimes incorrectly during field mapping (e.g. 'yyyy-dd-MM' instead of 'yyyy-MM-dd'). This scenario is best resolved within Movebank using option 2. Make sure the format string matches the format of your datetimes in the upper table exactly, click the button for 'Fixed offset from UTC' and make sure the 'Timezone offset' is set to 'UTC +0'.

Leave any non-Movebank supported fields unmapped (grey in the upper table).

When everything looks as you expect, hit the 'Finish' button at the bottom of the page.

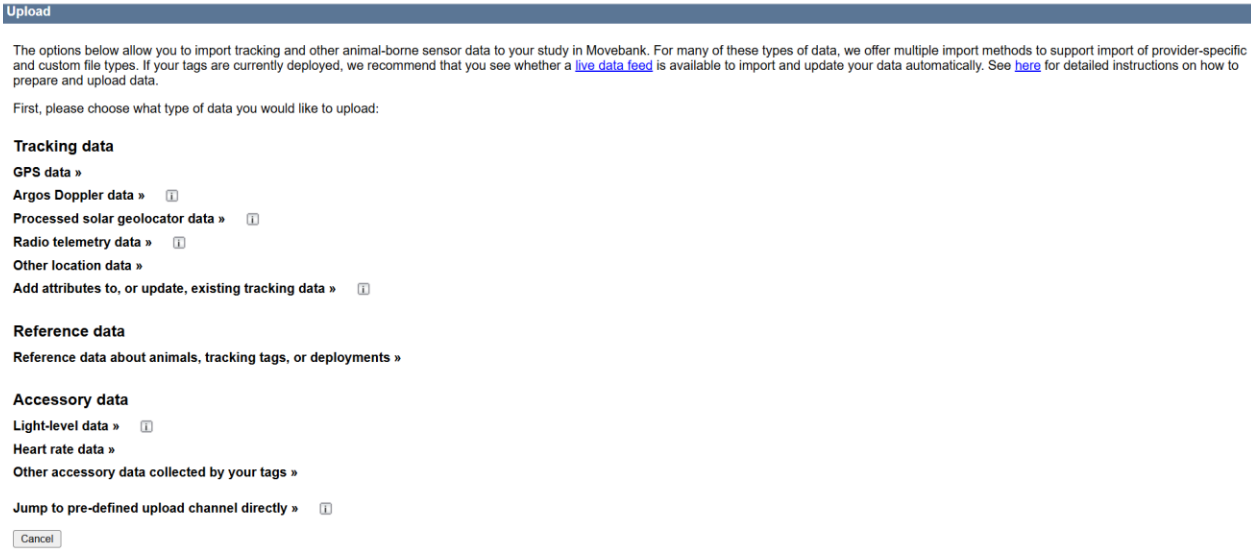
In the popup window, you'll be given the option to save a custom file format for later use (translation file to convert your data to Movebank's format). If you did not need to make any changes to Movebank's automatic field mapping, ignore this step and click 'ok'. If you did make changes and don't want to have to repeat them during subsequent uploads, check the box for 'Save file format copy as:' and replace the default file name with something meaningful, ideally including the date. This formatting file will always exist in your project. If you make further formatting changes, you will need to save a new custom file format and specify a new filename. Format files cannot be overwritten. Giving a new formatting file the same name as an existing one confusingly results in the two formatting options with the same name and no way to differentiate them!

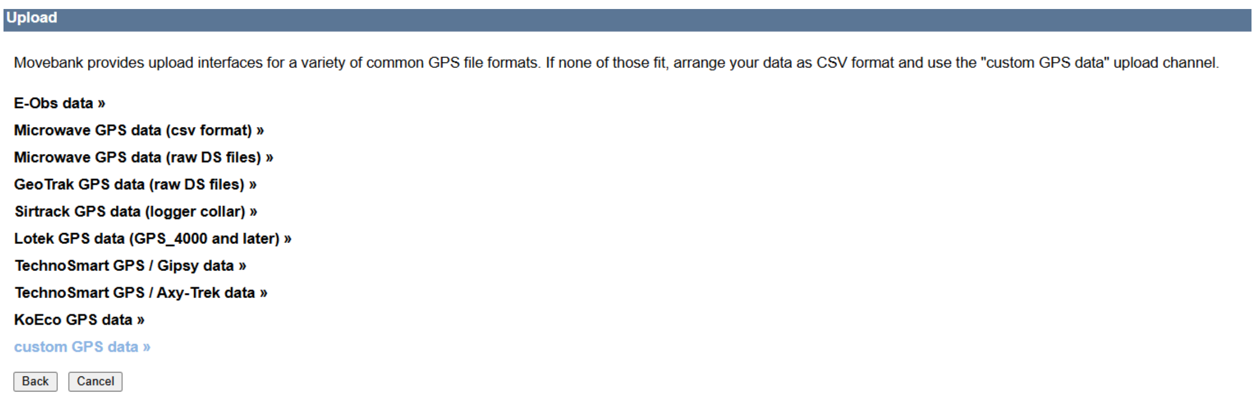
You should now see an 'Import successful' window. You can choose to return to the study main page or upload another file.

Event-level data

GPS positional data

Back on the upload landing page (Upload -> Import Data), select 'GPS data' under the heading 'Tracking data', then select 'custom GPS data'.





[If this is not your first time uploading custom GPS data to this study, a popup window will offer you the chance to select a custom file format from your saved options (a translation file to convert your data to Movebank's format). Select the format that is the best match for your upload so you don't need to manually map all your fields. You will still be able to make changes to the format as necessary. Do not select 'skip'.]

Navigate to your file using the 'Choose file' button, select it, then hit the upload button.

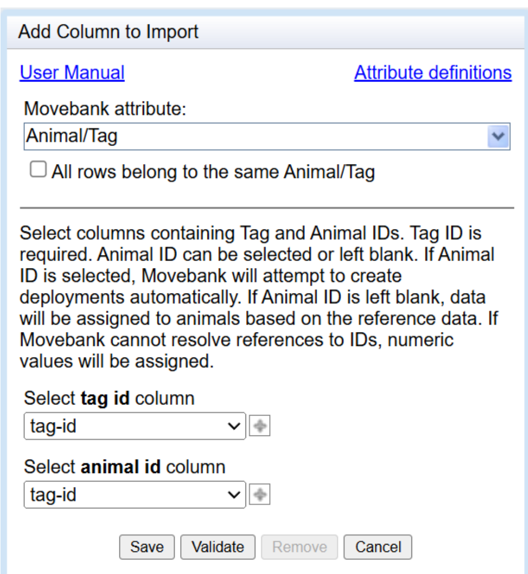
Click Ok to acknowledge the permissions popup window.

As with the reference data, the next page will show you two tables – the top table shows your dataset as uploaded and the bottom table shows your dataset as it will be saved in Movebank's database. However, **this time the lower table will be blank except for 'Sensor Type' = 'GPS'**. This is because we are using the 'custom' option so Movebank will not make any assumptions about your data format.

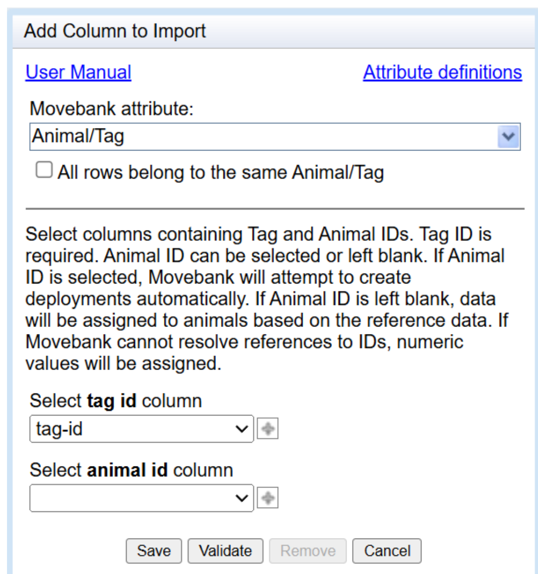
You will need to click on each heading in the upper table and map it to Movebank's accepted attributes. Leave any non-Movebank supported fields unmapped. When you click, a pop-up will appear where you can choose the attribute name and indicate what units were used in your import file. If you have already reformatted your field names and units to match those in Movebank's attribute dictionary, you likely will not need to change much, but please pay attention and ensure everything looks ok before hitting 'save' for that field – and especially before hitting 'finish' for the upload!

If you used a formatting function from seamoveR (or did not include an animal ID in your event-level data for some other reason), you will likely encounter Movebank trying to map your tag ID as both your 'tag-id' and your 'animal-id' when you click on the 'tag-id' column. You want Movebank to assign event-level data to animals based on the deployment and retrieval information in your reference data file, which requires toggling 'animal-id' to blank in the pop-up window as shown below. Do not miss this, or an additional animal ID will be added to your study and make things very confusing! (If you did include an animal ID in your event-level data, make sure the correct column is selected instead of leaving 'animal-id' blank.)

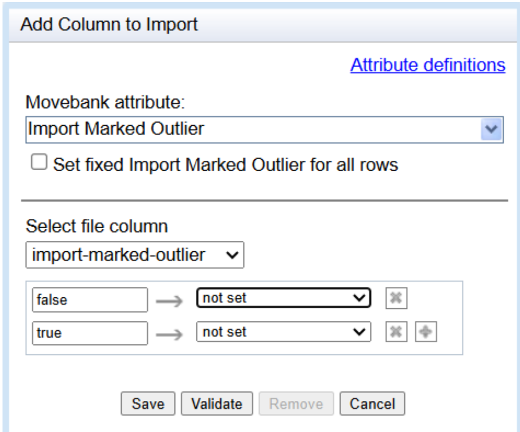
Incorrect:



Correct:



In addition, you will likely be required to match your categories for 'controlled list' variables to Movebank's categories. For example, even if your 'import marked outlier' field includes values 'false' and 'true', you may need to change 'not set' in the popup window to match the values in your column: 'false' and 'true'. If only one category appears and you expected several (Movebank only previews a limited number of rows looking for unique values), you will need to manually add the others using the '+' button. In this case, it's always good to double-check your data to make sure all expected values are actually present, instead of assuming the issue was caused by Movebank.



Finally, please check that the datetime columns are properly formatted (times are in UTC and match those in the upper table) and that latitude and longitude columns are not reversed.

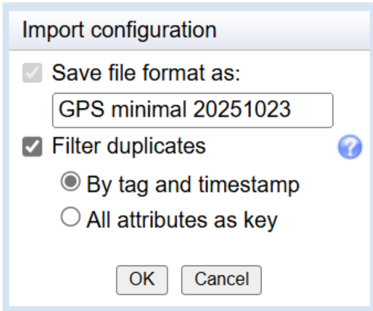
[See section 'Reference data' for how to handle other fields that do not map as expected.]

Click the Finish button.

An 'Import configuration' window will appear. If this is your first upload, it will force you to save your custom file format. Choose a meaningful name, ideally including the date.

[If this is not your first upload, you will be given the option to save a new custom file format. If you did not make changes to the automatic field mapping that you wish keep, leave the 'Save file format as:' checkbox blank. If you did make changes, select the box and replace the default filename with something meaningful, ideally including the date. This formatting file will always exist in your project and cannot be overwritten. Giving a new file the same name as an existing file confusingly results in the two formatting options with the same name and no way to differentiate them!]

You can also choose to filter duplicates at this step, either by tag ID and timestamp (removes rows from the same deployment that have identical timestamps) or using all imported attributes (removes rows that have identical values across all fields). It is typically fine to leave the default here, unless you're aware of a reason to change it. If you used a seamoveR formatting function, duplicates will already have been removed during the reformatting of event-level data.

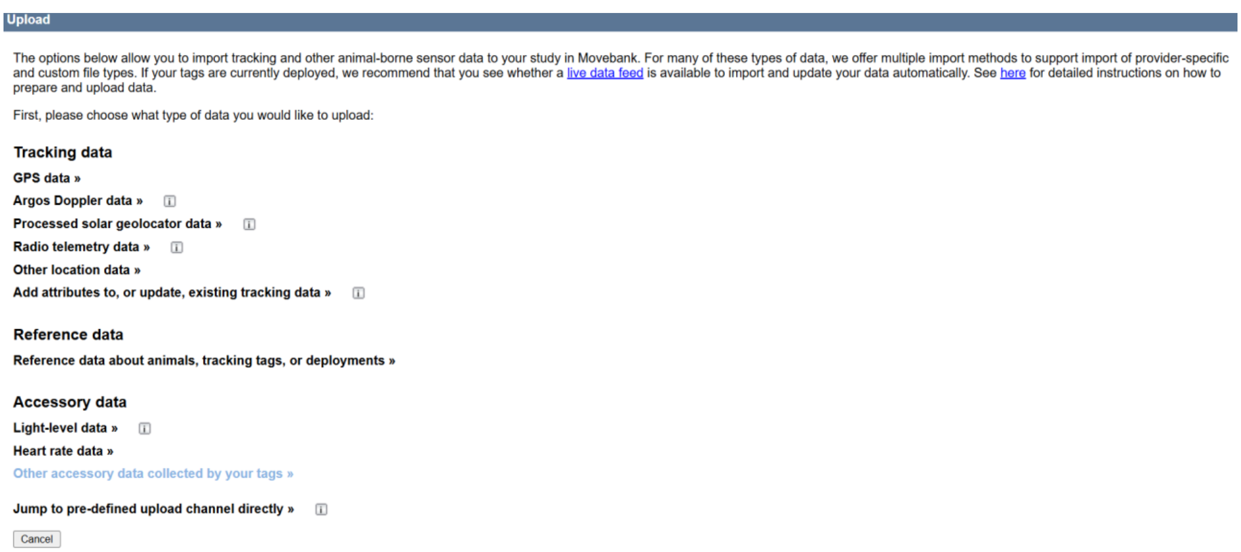


You should now see an 'Import successful' window. You can choose to return to the study main page or upload another file.

Accessory data

Please note: Movebank's database does not handle very large datasets well, so we don't recommend importing high-resolution accessory data such as accelerometer data – typically recorded at < 1 sec interval (> 1 Hz). These data should still be archived in the Attachments section of your study – ideally included within the 'input_data/event_data' subfolder folder of your zip file (see 'Prepare your data for Movebank' and 'Upload and import data to Movebank' -> 'Event-level data' -> 'Attachments'). If you have accessory data that are not high-resolution (e.g. TDR data), use the following instructions to import them.

Back on the upload landing page (Upload -> Import Data), select 'Other accessory data collected by your tags' under the heading 'Accessory data'.



[If this is not your first time uploading accessory data to this study, a popup window will offer you the chance to select a custom file format from your saved options (translation file to convert your data to Movebank's format). Select the format that is the best match for your upload so you don't need to manually map all your fields. You will still be able to make changes to the format as necessary. Do not select 'skip'.]

Navigate to your file using the 'Choose file' button, select it, then hit the upload button.

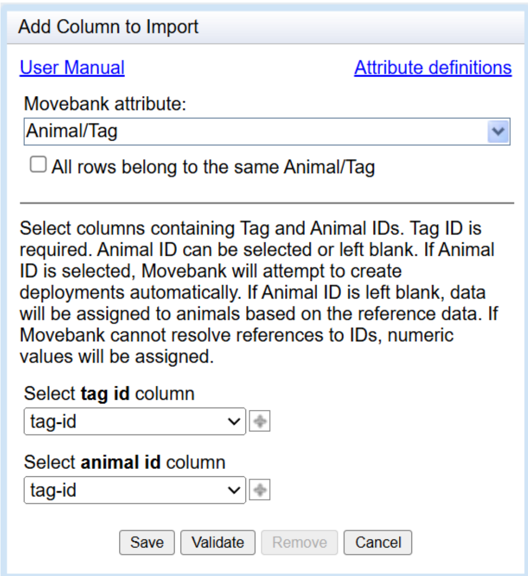
Click Ok to acknowledge the permissions popup window.

As with the reference and positional data, the next page will show you two tables – the top table shows your dataset as uploaded and the bottom table shows your dataset as it will be saved in Movebank's database. **The lower table will be blank except for 'Sensor Type' == 'Accessory Measurements'**. You should change this to a more specific sensor type by clicking on the field heading in the lower table and selecting another option from the 'Set value' dropdown menu, such as 'TDR'.

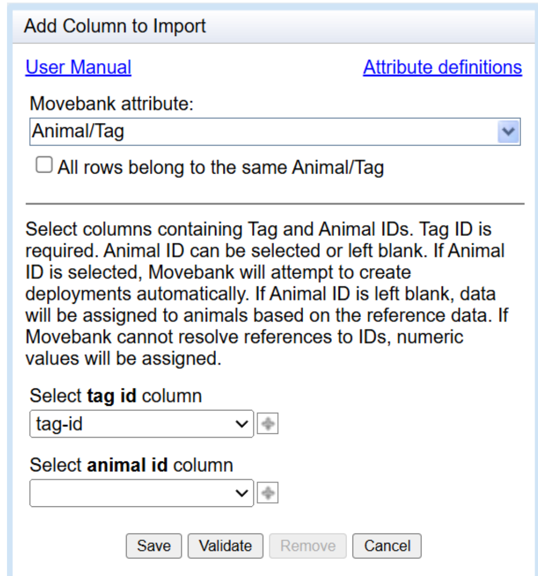
You will need to click on each heading in the upper table and map it to Movebank's accepted attributes. Leave any non-Movebank supported fields unmapped. When you click, a pop-up will appear where you can choose the attribute name and indicate what units were used in your import file. If you have already reformatted your field names and units to match those in Movebank's attribute dictionary, you likely will not need to change much, but please pay attention and ensure everything looks ok before hitting 'save' for that field – and especially before hitting 'finish' for the upload!

If you used a formatting function from seamoveR (or did not include an animal ID in your event-level data for some other reason), you will likely encounter Movebank trying to map your tag ID as both your 'tag-id' and your 'animal-id' when you click on the 'tag-id' column. You want Movebank to assign event-level data to animals based on the deployment and retrieval information in your reference data file, which requires toggling 'animal-id' to blank in the pop-up window as shown below. Do not miss this, or an additional animal ID will be added to your study and make things very confusing! (If you did include an animal ID in your event-level data, make sure the correct column is selected instead of leaving 'animal-id' blank.)

Incorrect:

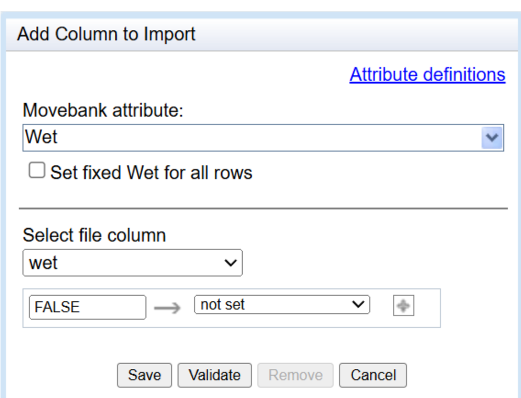


Correct:

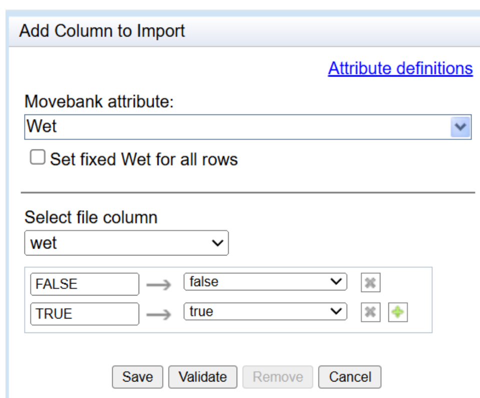


In addition, you will likely be required to match your categories for 'controlled list' variables to Movebank's categories. For example, even if your 'wet' field includes values 'false' and 'true', you may need to change 'not set' in the popup window to match the values in your column: 'false' and 'true'. If only one category appears and you expected several (Movebank only previews a limited number of rows looking for unique values), you will need to manually add the others using the '+' button. In this case, it's always good to double-check your data to make sure all expected values are actually present, instead of assuming the issue was caused by Movebank.

Incorrect:



Correct:



Finally, please check that the datetime columns are properly formatted (times are in UTC and match those in the upper table) and that latitude and longitude columns are not reversed.

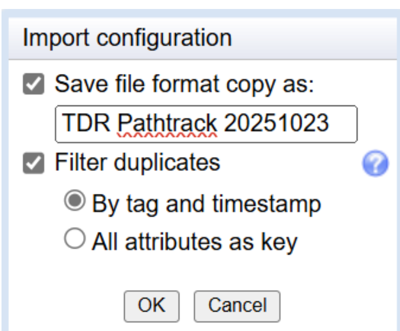
[See section 'Reference data' for how to handle fields that do not map as expected.]

Click the Finish button.

An 'Import configuration' window will appear. If this is your first upload, it will force you to save your custom file format. Choose a meaningful name, ideally including the date.

[If this is not your first upload, you will be given the option to save a new custom file format. If you did not make changes to the automatic field mapping that you wish keep, leave the 'Save file format as:' checkbox blank. If you did make changes, select the box and replace the default filename with something meaningful, ideally including the date. This formatting file will always exist in your project and cannot be overwritten. Giving a new file the same name as an existing file confusingly results in the two formatting options with the same name and no way to differentiate them.]

You can also choose to filter duplicates, either by tag ID and timestamp (removes rows from the same deployment that have identical timestamps) or using all imported attributes (removes rows that have identical values across all fields). It's typically fine to leave the default here, unless you're aware of a reason to change it. If you used a seamoveR formatting function, duplicates will already have been removed during the reformatting of event-level data.



You should now see an 'Import successful' window. You can choose to return to the study main page or upload another file.

Validation

Movebank has an overview of basic quality control steps here.

From the Studies landing page, you should review the **'Study Statistics'** section to ensure you see the number of deployments, locations, and outliers that you expect. It can take some time (several hours) for the changes you made to be reflected correctly in the Study Statistics. Once the 'Processing status' at the bottom of the page has changed from 'Map &

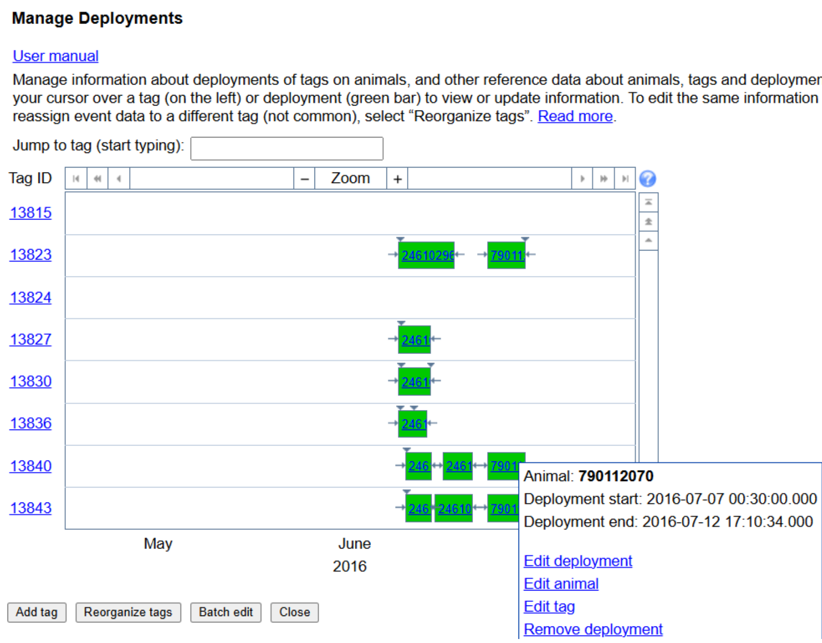
study statistics processing' to 'Up-to-date', you can proceed with validation.

Study Statistics	
Number of Animals	144
Number of Tags	117
Number of Deployments	176
Time of First Deployed Location	2016-06-23 00:30:41.000
Time of Last Deployed Location	2023-09-08 19:30:18.000
Taxa	Oceanodroma leucorhoa
Number of Deployed Locations	16561
Number of Records	Deployed (outliers) / Total (outliers)
GPS	16561 (6) / 16561 (6)
About study details	
Processing Status	Up-to-date

You can also click the 'View' button on the navigation bar at the top of the studies page, then 'Show in map' to check that your tracks appear as expected.

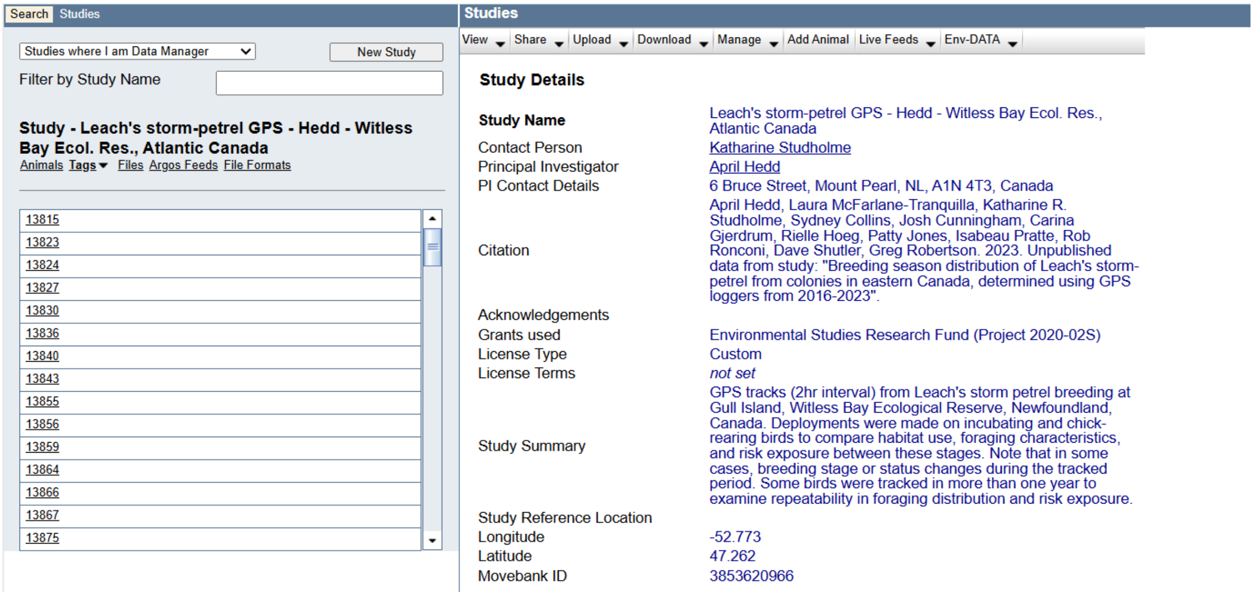
Movebank also provides several interactive review and editing options, accessed via the 'Manage' button in the navigation bar. While we don't recommend making any substantive changes to your data here (corrections should be made reproducibly within your R script), these can be good tools for detecting errors.

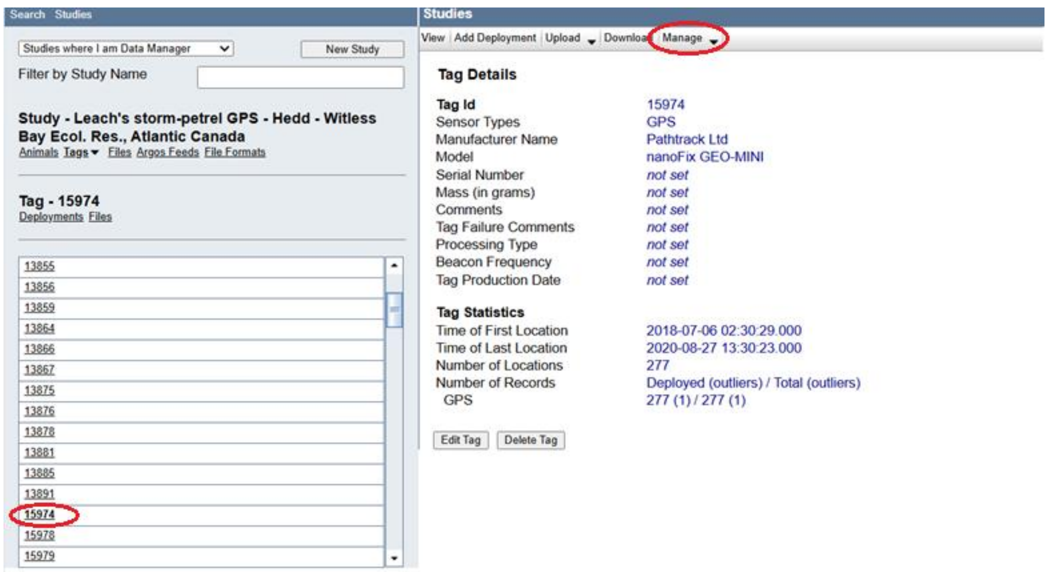
The **'Deployment Manager'** displays a timeline chart with tag ID on the y-axis and date on the x-axis. Small downward-pointing arrows mark the first and last data point from each tag, and green bars span the dates during which the tag was deployed. The animal-id (band number) for each deployment is displayed on each bar. Hovering your mouse over the bar will show a pop-up with the exact deployment timestamps. This plot can help identify incorrect deployment timestamps (e.g. deployment-on-date prior to when data actually exist).

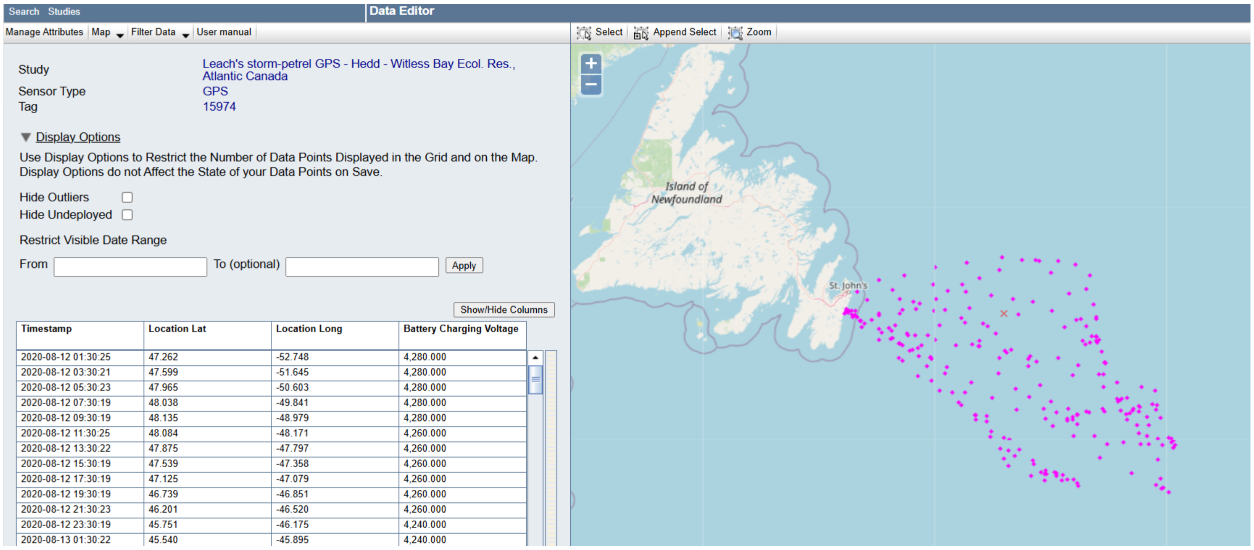


The **'Event Editor'** will display a map with positions as pink diamonds (similar to 'View' -> 'Show in map'), but also displays the points labelled as outliers as red 'x's and all timestamps and locations in a side panel. This tool could be used to manually identify extreme outliers if you wish (but ONLY extreme outliers, see section 'Pre-process positions'). For guidance on how to do this, see this section of Movebank's user manual.

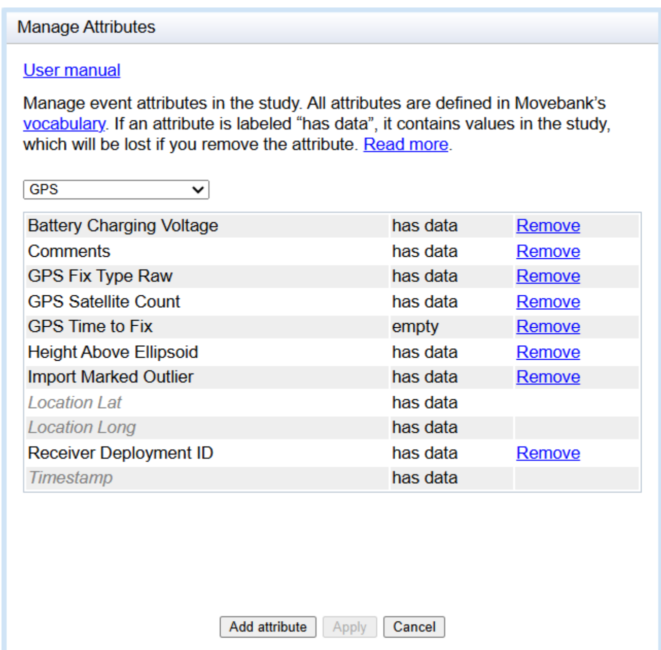
You can view the event editor for one tag at a time if you select a single tag BEFORE entering the Event Editor. To do this, click 'Tags' under the study name in the left-hand pane of main study page, then click your tag of choice, then click 'Manage' -> 'Event Editor' from the main navigation bar in the right-hand pane. Once there, clicking on the attribute table will draw arrows between all points within that deployment and highlight the point corresponding to the row you clicked.


Search Studies
Studies
View Share Upload Download Manage Add Animal Live Feeds Env-DATA
Studies where I am Data Manager
New Study
Filter by Study Name
Study - Leach's storm-petrel GPS - Hedd - Witless Bay Ecol. Res., Atlantic Canada
Animals Tags Files Argos Feeds File Formats
13815
13823
13824
13827
13830
13836
13840
13843
13855
13856
13859
13864
13866
13867
13875
Study Details
Study Name
Leach's storm-petrel GPS - Hedd - Witless Bay Ecol. Res., Atlantic Canada
Contact Person
Katharine Studholme
Principal Investigator
April Hedd
PI Contact Details
6 Bruce Street, Mount Pearl, NL, A1N 4T3, Canada
Citation
April Hedd, Laura McFarlane-Tranquilla, Katharine R. Studholme, Sydney Collins, Josh Cunningham, Carina Gjerdrum, Rielle Hoeg, Patty Jones, Isabeau Pratte, Rob Ronconi, Dave Shutler, Greg Robertson. 2023. Unpublished data from study: "Breeding season distribution of Leach's storm-petrel from colonies in eastern Canada, determined using GPS loggers from 2016-2023".
Acknowledgements
Grants used
Environmental Studies Research Fund (Project 2020-02S)
License Type
Custom
License Terms
not set
Study Summary
GPS tracks (2hr interval) from Leach's storm petrel breeding at Gull Island, Witless Bay Ecological Reserve, Newfoundland, Canada. Deployments were made on incubating and chick-rearing birds to compare habitat use, foraging characteristics, and risk exposure between these stages. Note that in some cases, breeding stage or status changes during the tracked period. Some birds were tracked in more than one year to examine repeatability in foraging distribution and risk exposure.
Study Reference Location
Longitude
-52.773
Latitude
47.262
Movebank ID
3853620966


Search Studies
Studies
View Add Deployment Upload Download Manage
Studies where I am Data Manager
New Study
Filter by Study Name
Study - Leach's storm-petrel GPS - Hedd - Witless Bay Ecol. Res., Atlantic Canada
Animals Tags Files Argos Feeds File Formats
Tag - 15974
Deployments Files
13855
13856
13859
13864
13866
13867
13875
13876
13878
13881
13885
13891
15974
15978
15979
Tag Details
Tag Id
15974
Sensor Types
GPS
Manufacturer Name
Pathtrack Ltd
Model
nanoFix GEO-MINI
Serial Number
not set
Mass (in grams)
not set
Comments
not set
Tag Failure Comments
not set
Processing Type
not set
Beacon Frequency
not set
Tag Production Date
not set
Tag Statistics
Time of First Location
2018-07-06 02:30:29.000
Time of Last Location
2020-08-27 13:30:23.000
Number of Locations
277
Number of Records
Deployed (outliers) / Total (outliers)
GPS
277 (1) / 277 (1)
Edit Tag
Delete Tag



The **'Attribute Manager'** can be used to see a list of the fields you have in your position or accessory datasets and whether those fields contain values. This can be a nice double-check to make sure you have all the fields you intended.



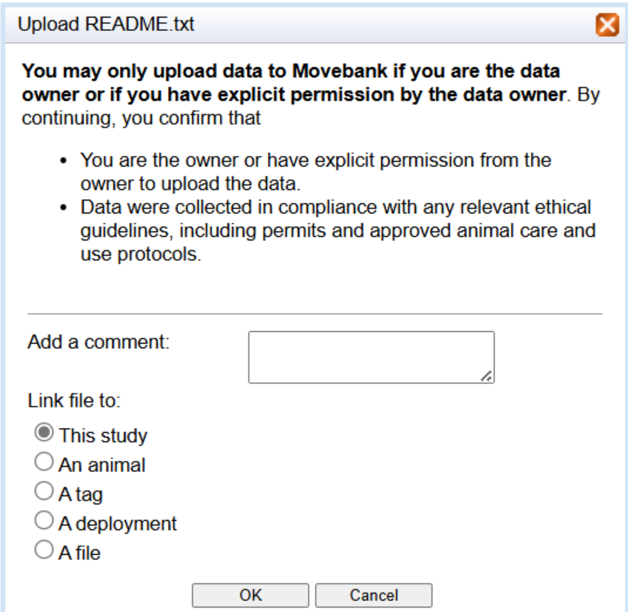
Attachments

Once you are done working on your study for a while, please upload the following files to the Attachments section:

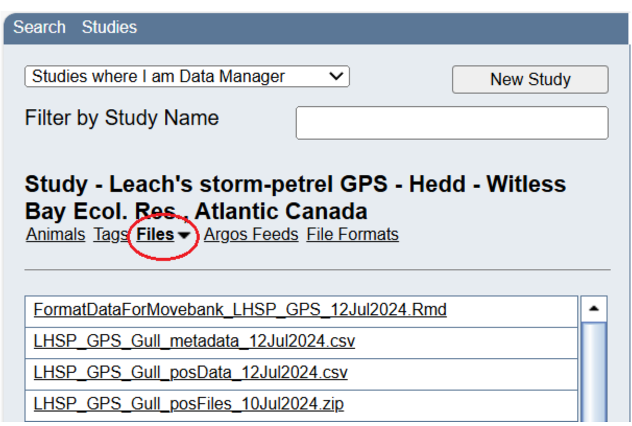
- Zipped folder containing original data files and file processing scripts 'MovebankDataPrep'_spcd_tagType_site_yyyymmdd.zip

- User guidance file
  README.txt

On the main study page, select 'Upload' -> 'Add Attachments', then 'Choose file' to navigate to and upload your file. A popup window will appear. Leave 'Link this file to' set to 'This study', add a comment if you like, and click 'OK'.



You can view all the files attached to your study by clicking 'Files' under the study name in the left-hand pane of the study's landing page. These will include files you've added using 'Add Attachments' as well as files automatically added to the Attachments section by Movebank (original reference data and event-level data files). These files can be downloaded, deleted, or overwritten/replaced anytime. For example, if you are using part of your README.txt as a to-do list for the study, you can replace the existing file with a new version each time you work on the study (see section 'Modifications/Corrections').



Study Modifications or Annual Update

If you need to modify the data in your study for any reason, or the time has come to add another year of data to your project, you should complete all changes within the R project for your study. For example, if you are adding a new year of deployments and event-level data, you would:

1. Save the new year's reference data file in 'spcd-tagType-site/input_data/ref_data',
2. Append the new reference data to your existing reference data in Excel or in R and save it in 'spcd-tagType-site/output_data',
3. Add your newly downloaded tag files alongside your old ones in 'spcd-tagType-site/input_data/event_data')
4. Re-run your data reformatting scripts so the files for Movebank upload are regenerated (including both the old and new tag data) in 'spcd-tagType-site/output_data',
5. Re-upload these revised files to Movebank, including field mapping steps if applicable (see instructions for each below), and
6. Update, re-generate, and re-upload the user guidance (README) file

Having a copy of the file(s) you will be updating temporarily backed up on your computer can be helpful for troubleshooting if your revisions don't go as planned. You can download the current version of each file from Movebank by clicking 'Files' under the study name in the left-hand pane of the study's landing page and then clicking on the filename of interest. A 'File Details' page will appear in the right-hand pane. In the grey navigation bar at the top of this pane, click 'Download' -> 'Download original file'.

Reference data

When your revised file is ready, initiate the standard upload process ('Upload' -> 'Import Data' -> 'Reference data about animals, tracking tags, or deployments' -> 'Use Movebank standard reference data format').

Since the new file name will be the same as the original one, the popup window that appears after you click 'Upload' will have a new section indicating that: a) the study already contains deployments and b) a reference file of the same name has already been uploaded. A checkbox will automatically be selected next to each warning, indicating that the original deployments in the Movebank database and your original reference data file will be replaced by this new upload. Click 'OK' and follow the standard upload process outlined in section 'Upload and import data to Movebank' -> 'Reference data'.

Event-level data

As with reference data, the new event-level data file name will be the same as the original one; however, but if you were to upload this file it would be stored as a *second* event-level data file with the file name appended with '_002'. This is not the desired outcome – we want to avoid accumulating multiple versions of files in the study. If you see a pop-up warning that this will occur, click 'Cancel'.

Before you upload your updated event-level data, you will need to **delete your old event-level data from Movebank**. We recommend that you download (back up) the original event-level data file before doing this as described at the start of this section.

To delete a file, navigate to the 'File Details' page (see description at the start of this section). At the bottom of this page, click the 'Delete file' button and then 'Ok'.

When your revised file is ready, initiate the standard upload process for event-level data ('Upload' -> 'Import Data' -> [Data Type]).

This event-level data replacement process also ensures that all years of data are processed in exactly the same way, rather than risking having slightly different processing steps used in different years.

Data preparation folder

If your modifications resulted in changes to your input data or R scripts, you need to update Movebank's copy of your data preparation zip folder. Recreate this zip folder from the most recent version of your R project (see section 'Prepare your data for Movebank') and name it with the current date: 'MovebankDataPrep'_spcd_tagType_site_yyyymmdd.zip

Initiate the standard upload process for attachments ('Upload' -> 'Add attachments'). Unlike reference data and event-level data files, your new zip folder will not match the filename already on Movebank. You will not see any warnings, and you will end up with two versions of your data preparation folder in your Attachments. **Delete the outdated version.**

User guidance file

When your revised README file is ready, initiate the standard upload process for attachments ('Upload' -> 'Add attachments'). Once you have navigated to your new file and clicked upload, the popup window that appears will indicate that a file of the same name has already been uploaded. A checkbox will automatically be selected, indicating that the original file will be replaced by the new upload. This is the desired outcome. Click 'OK'.

More help please?

Please contact katharine.studholme@ec.gc.ca or WRD Tracking East and West with questions and comments. For more detailed instructions and troubleshooting, please first refer to Movebank's User Manual. Don't forget that we have our own ECCC Movebank Attribute Dictionary that should be referenced in place of the one on the Movebank website!